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RESEARCH AND EDUCATION

Deep learning-based automated detection of the dental crown finish line: An accuracy study

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ABSTRACT

Statement of problem. The marginal fit of dental prostheses is a clinically significant issue, and dental computer-aided design software programs use automated methods to expedite the extraction of finish lines. The accuracy of these automated methods should be evaluated.

Purpose. The purpose of this study was to compare the accuracy of a new hybrid method with existing software programs that extract finish lines using fully automated and semiautomated methods.

Material and methods. A total of 182 jaw scans containing at least 1 natural tooth abutment were collected and divided into 2 groups depending on how the digital data were created. Group DS used desktop scanners to scan casts trimmed for improved finish line visibility, while Group IS used intraoral scans. The method from Dentbird was compared using 3 software packages from 3Shape, exocad, and MEDIT. The Hausdorff and Chamfer distances were used in this study. Three dental laboratory technicians experienced in the digital workflow evaluated clinical finish line acceptance and its Hausdorff and Chamfer distances. For statistical analysis, *t* tests were performed after the outliers had been removed using the Tukey interquartile range method (α=.05).

Results. Outliers identified by using the Tukey interquartile range method were more numerous in the semiautomatic methods than in the automatic methods. When considering data without outliers, the software performance was found to be similar for desktop scans of the trimmed casts. However, the method from Dentbird demonstrated statistically better results (P<.05) for the posterior tooth with finish lines in concave regions than the 3Shape, exocad, and MEDIT software programs. Furthermore, thresholds coherent with clinical acceptance were determined for the Hausdorff and Chamfer distances. The Hausdorff distance threshold was 0.366 mm for desktop scans and 0.566 mm for intraoral scans. For the Chamfer distance, the threshold was 0.026 for desktop scans and 0.100 for intraoral scans.

Conclusions. The method from Dentbird demonstrated a comparable or better performance than the other software solutions, particularly excelling in finish line extraction for intraoral scans. Using a hybrid method combining deep learning and computer-aided design approaches enables the robust and accurate extraction of finish lines. (J Prosthet Dent 2024;132:1286.e1-e9)

Marginal fit in dentistry refers to the accuracy of the fit between a prosthesis and the prepared tooth structure, and to how well the prosthesis fits along the margin of the restoration. Accurate marginal fit is essential, as it influences the restoration's clinical longevity and performance. A precise marginal fit helps prevent microgaps where bacteria can accumulate, reducing the risk of caries and ensuring the restoration remains functional and comfortable.[1–4] The accurate placement of the finish lines, which mark the junction of prepared and unprepared tooth structure and the margin of a restorative material, is essential.

In the traditional approach, the finish line is drawn on a stone cast, but computer-aided design (CAD) technology has become popular and offers advantages in accuracy, convenience, and repeatability.[3]

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Clinical Implications

The hybrid method demonstrated comparable or better accuracy in finish line extraction, suggesting its potential for enhancing marginal fit. Additionally, the study provided thresholds for the Hausdorff and Chamfer distances, enabling clinicians to assess the feasibility of finish lines. This information facilitates evidence-based decision-making when selecting a software program, ultimately leading to improved outcomes, reduced rework, and increased patient satisfaction.

Nonetheless, drawing a 3-dimensional (3D) curve using CAD can be a time-consuming and challenging task.[5] To facilitate finish line placement, automated methods have been developed based on CAD or deep learning techniques.

CAD-based methods[3,6–8] rely on rule-based techniques to extract boundary curves. First, a morphological skeleton algorithm is used for tooth segmentation.[7,9] The initial step involves assigning a category to vertices, followed by the gradual removal of the outermost layer. The assigned category is used to separate the mesh into 2 regions, identifying the seed point's region and extracting the tooth accordingly.[7] Second, a harmonic field is used for tooth boundary extraction.[8,10] This method uses feature points to create a harmonic field, which is used to extract a boundary curve.[8] To extract finish lines, Shin et al[3] used a region-growing technique[11–13] with a dynamic weight function[14–16] provided with a seed point. However, fully automated CAD methods for extracting tooth boundaries are lacking. Additionally, existing research highlights the limitations of study robustness.[3,7,8]

In contrast, deep learning-based methods[4,17–27] use deep neural networks trained on large datasets. Cui et al[19] proposed TSegNet, which detects all teeth using a distance-aware tooth centroid voting scheme followed by a confidence-aware cascade. Lian et al[26] proposed MeshSegNet, which predicts tooth boundaries[26] and anatomic landmarks[27] using graph-cut-refinement.[28] Although deep learning methods are effective, they have practical limitations, including that these models cannot handle patients who deviate from the typical arch shape. Additionally, the postprocessing step is dependent on sensitive parameter values, which can affect overall accuracy.[26,27] To address these issues, a hybrid method that combines the strengths of CAD-based and deep learning-based methods can be used.

This study aimed to test the research hypothesis that a significant difference would be found in the accuracy of the finish lines obtained through the hybrid method used by Dentbird compared with the CAD-based method provided by 3Shape, exocad, and MEDIT. Additionally, this study sought to establish a connection between quantitative metrics and clinical feasibility, thus supporting the validity of the assessment.

MATERIAL AND METHODS

This study used retrospective convenience sampling and did not involve patient recruitment for data collection. The data evaluated in this study were a set of anonymized scans from different clinics that had been originally collected for clinical purposes. The Seoul National University Institutional Review Board (IRB) approved the study (IRB number: ERI21005). A total of 182 jaw scans containing at least 1 natural tooth abutment were collected and divided into 2 groups depending on how the digital data were created (Fig. 1). Group DS included the scanned data of casts trimmed for improved finish line visibility from a desktop scanner, including 58 anterior and 83 posterior teeth. Group IS had intraoral scan data, including 35 anterior and 58 posterior teeth.

The accuracy of the hybrid method (Dentbird Crown 2.1.4; Imagoworks Inc) was evaluated by comparing it with 3 other software programs: Dental System 2021–1 2.21.2.1 (3Shape A/S), DentalCAD 3.1 Rijeka (exocad), and Medit Margin Lines 1.0.0.32 (MEDIT Corp). The experiment involved evaluating the accuracy of 4 automated programs by comparing their results with the reference finish lines manually drawn by 3 dental laboratory technicians with more than 5 years of experience.

The Dentbird software program used deep learning-based and CAD-based algorithms: the preparation tooth detector, finish line segmentor, and spline curve theory

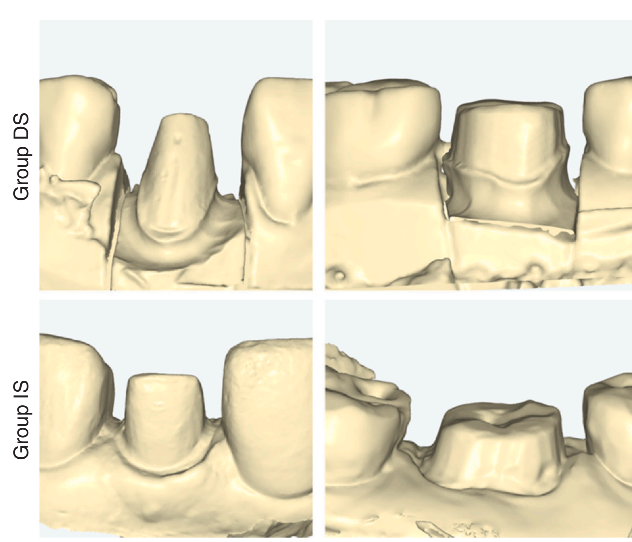

Figure 1. Data categorized into two groups depending on how digital data created: scans from desktop scan (Group DS) shown in upper row, and scans from intraoral scan (Group IS) presented in lower row.

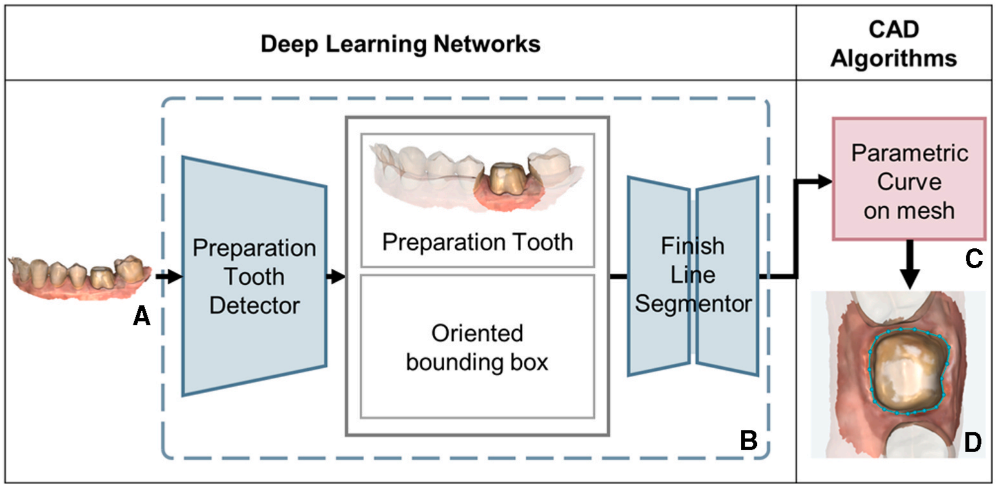

Figure 2. Overview of hybrid method used by Dentbird solution. A, Image of scan of patient's natural tooth abutment. B, Deep learning networks. C, Parametric curve-on-mesh algorithm. D, Obtaining optimal curve for preparation finish line.

(Fig. 2). The preparation tooth detector is a convolutional neural network (CNN)-based model[29] that takes scan data as input to predict the position, orientation, and tooth number for the preparation of the prosthesis. The model was built upon CenterNet[30] and used the Stacked Hourglass Networks as its backbone architecture.[31] After extracting the feature maps from the backbone, the model used 3 separate branches. Each branch predicted the arch type, mesial directions, and tooth types with sizes. The training procedure adopted multitasking learning to exploit shared prediction among branches. The finish line segmentor used a CNN-based encoder-decoder architecture, named UneXt,[32] to generate a finish line represented by a set of points. It took a parametrized image from input mesh, and the boundary of the region was extracted as discrete points and transformed into a smooth parametric curve using spline curve theory.[33] A curve-on-mesh technique was used to ensure the curve lines on the surface.

To generate the finish line in the Dental System software program, the manual provided by this software vendor was followed.[34] Different workflows were used depending on how the scan data were obtained. For the intraoral scan, the system automatically generated a finish line. If the automatic extraction failed, the system offered 2 options. To advance to the next step, the option that aligned with the data was chosen. The first option required a point to be marked on the occlusal surface of the abutment, whereas the second option required a point to be marked on the finish line. For the desktop-scanned data, the user was asked to select a point of finish lines for each natural tooth abutment. To evaluate desktop scans (Group DS), the model option was used as the data type, and a point on the finish line was selected. For intraoral scans (Group IS), the digital scanning option was used as the data type. If the system failed to segment the finish line, the option with a point on the natural tooth abutment was used, as it offered more consistent results than one requiring a finish line point.

The DentalCAD software program had 2 finish line options: supragingival and subgingival. The supragingival option is used to detect finish lines on convex surfaces and requires a point on the finish line. The subgingival option is used to detect finish lines on concave surfaces and requires at least 4 points. The supragingival option was used in Group DS, and the subgingival option in Group IS. For Group DS, a single point was marked on the finish line, whereas for Group IS, 4 points were marked, namely from the buccal, mesial, lingual, and distal sides. If the finish line was inaccurate, the best results from the DentalCAD software program were attempted at least 3 times with different inputs.

The Medit software program required data obtained from its own scanning process and a point on the finish line to generate a finish line. The same workflow was used for Groups DS and IS. The experimental dataset was obtained by a 4-step process. First, the data were attached to a new patient. Next, a pseudoscanning process was performed. Third, a margin extension application was launched. Lastly, a point on the finish line was clicked. The best results from the Medit software program's automatic finish line detection were attempted at least 3 times with different input points.

The Hausdorff distance (HD), a metric aimed to measure the similarity between 2 closed curves can be applied to a discrete domain.[35,36] To determine 1-way HD from the reference set X and the extracted set Y, the maximum distance between each point in X and its closest point in Y was calculated: $d_H(X, Y)=\max_{x\in X}(\min_{y\in Y}\|x-y\|)$, and the 2-way HD was used from: $D_H(X, Y)=\max(d_H(X, Y), d_H(Y, X))$. The chamfer distance (CD) is a metric to evaluate the similarity between point clouds.[37,38] To mitigate the sensitivity to outliers and emphasize the differences between the measured and reference values, the squared L2 norm for CD measurements was used as follows:

$$D_C(X, Y) = \frac{1}{n_x}\sum_{x\in X}\min_{y\in Y}\|x-y\|_2^2 + \frac{1}{n_y}\sum_{y\in Y}\min_{x\in X}\|x-y\|_2^2$$

A comprehensive evaluation was conducted by 3 experienced dental laboratory technicians, each with more than 5 years of experience in digital dentistry, to establish an association between the clinical acceptance of the finish line and its HD and CD values. They all had extensive experience with all the software programs. The results of the assessment were validated through random sampling during interexamination. Threshold values for each group and metrics to indicate the clinical acceptance of the finish lines were established.

The threshold that aligned with the dental laboratory technicians' assessment was computed. They distinguished the labels $L = \{l_1,...,l_N\}$ given the finish lines $M = \{m_1, ..., m_N\}$. l_i is 1 if m_i fulfilled the following criteria: first, the finish line should prevent exposure to dentin; secondly, the finish line should avoid exerting pressure on the gingiva; lastly, the finish line should not invade the gingiva or adjacent tooth region; otherwise l_i is 0. Given a threshold t, 2 sets were defined: $M_1(t)$, which included indices of metric values v_i below t, and $M_2(t)$, which included indices of v_i above or equal to t. The threshold, τ, that maximized classification accuracy was computed as follows:

$$\tau = argmax_t \frac{|\{i: i \in M_1(t)\ and\ l_i = 1\}| + |\{i: i \in M_2(t)\ and\ l_i = 0\}|}{N}$$

t tests were performed to compare HD and CD between the types of software programs. Before evaluating P values, outliers were removed by using the Tukey interquartile range (IQR) method.[39] The quartiles of the HD values were computed to obtain the IQR. The threshold was defined as 1.5 times the IQR. The outliers were excluded from the statistical analysis.

RESULTS

The outlier classification results obtained by using the Tukey IQR method are shown in Table 1. The results for each group are described in the text and shown in Figures 3 and 4, and representative preparations to illustrate the characteristics of HD and CD measurements are shown in Figures 5 and 6.

The results for Group DS were as follows: for the anterior teeth, no significant differences were found in the HD and CD (P>.05), whereas, for the posterior teeth, significant differences were observed in the HD and CD between the Medit software program and the other software packages (P<.05) (Table 2). Regarding the posterior teeth, the Medit software program had the largest mean value for all metrics (Table 3). The results for clinical acceptance indicated that of the 340 preparations assessed, 305 were considered acceptable. The established thresholds for HD and CD were 0.366 mm and 0.026, respectively (Table 4). The DentalCAD software program showed the highest number of preparations meeting the thresholds, with a total of 79, followed by the Dentbird software program with 76, the Dental System software program with 74, and the Medit software program with the lowest count of 70 preparations.

For Group IS, significant differences were observed for the anterior teeth in terms of the HD between the Dentbird software program and the Dental System software program and between the Dental System software program and the DentalCAD software program, and in the CD between the Dentbird software program and the Dental System software program (P<.05). For the posterior teeth, significant differences were found in terms of the HD and CD between the Dentbird software program and the others (P<.05) (Table 2). Regarding the posterior teeth, the Dentbird software program had the smallest mean values for all metrics (Table 3). The findings from the clinical acceptance evaluation showed that of the 332 preparations examined, 194 were considered acceptable. The determined thresholds for HD and CD were 0.566 mm and 0.100, respectively (Table 4). The Dentbird software program exhibited the highest count with 54 preparations meeting the thresholds, followed by the DentalCAD software program with 44 preparations. Both the Dental System software and the Medit software program had the lowest count, each with 37 preparations meeting the thresholds.

DISCUSSION

The null hypothesis of the study, that the finish lines from the hybrid method in the Dentbird software program would

Table 1. Statistics of outliers exceeding 1.5 interquartile range value of Hausdorff distance

	Tooth Type	Total Preparations	1.5 IQR HD [mm]	Outliers				
				Total	Dentbird	Dental System	DentalCAD	Medit
Group DS	Anterior	58	0.814	25 (43%)	2 (3%)	22 (38%)	2 (3%)	5 (9%)
	Posterior	83	0.715	31 (37%)	3 (4%)	22 (27%)	9 (11%)	13 (16%)
Group IS	Anterior	35	1.215	3 (9%)	2 (6%)	0 (0%)	0 (0%)	2 (6%)
	Posterior	58	1.658	7 (12%)	1 (2%)	3 (5%)	2 (3%)	4 (7%)

HD, Hausdorff distance; IQR, interquartile range.

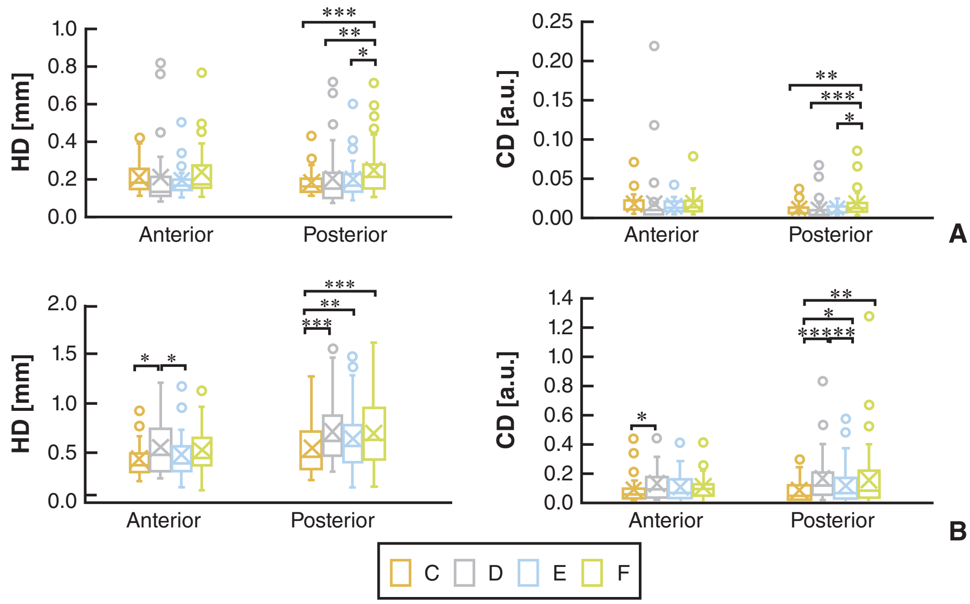

Figure 3. Boxplot denoting statistical significances (*P<.05; **P<.01; ***P<.001) within 1.5 interquartile range value of Hausdorff distance and Chamfer distance. A, Average results on 33 anterior teeth and 52 posterior teeth in desktop scan (Group DS), respectively. B, Average results on 32 anterior teeth and 51 posterior teeth in intraoral scan (Group IS), respectively. C, Dentbird software program. D, Dental System software program. E, DentalCAD software program. F, Medit software program. CD, Chamfer distance. HD, Hausdorff distance.

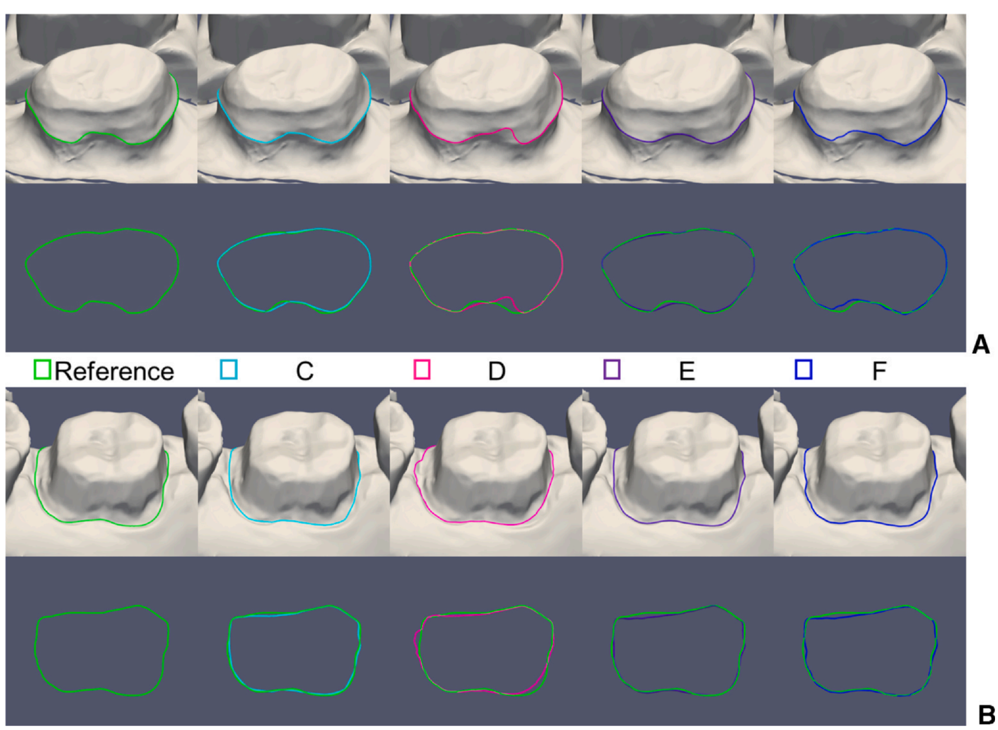

Figure 4. Comparison between reference and finish lines produced by each software program. A, Example of finish lines for desktop scan (Group DS). B, Example of finish lines for intraoral scan (Group IS). C, Dentbird software program. D, Dental System software program. E, DentalCAD software program. F, Medit software program.

be more accurate than those from the software programs provided by 3Shape, exocad, and MEDIT, was not supported. However, the statistical analysis indicates that the hybrid method yielded better results than the Medit software program for posterior teeth in Group DS and outperformed the DentalCAD software program for anterior

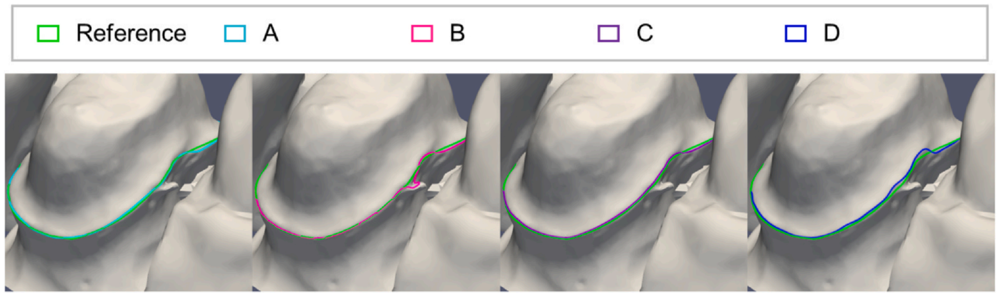

Figure 5. Representative preparation illustrating characteristics of metrics: HD ordering A (0.159)<C (0.160)<D (0.239)<B (0.384); CD ordering à A (0.008)<B (0.010)<C (0.011)<D (0.014). A, Dentbird software program. B, Dental System software program. C, DentalCAD software program. D, Medit software program. CD, Chamfer distance. HD, Hausdorff distance.

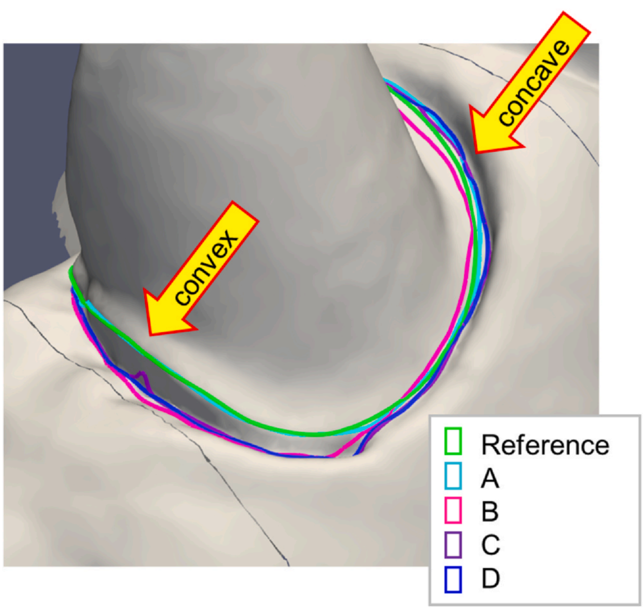

Figure 6. Complex convexity preparation. Dentbird software program robust for both concave and convex regions, while other software programs fail to extract convex regions. A, Dentbird software program. B, Dental System software program. C, DentalCAD software program. D, Medit software program.

teeth in Group IS. Furthermore, the data suggested that for posterior teeth in Group IS, the Dentbird software program demonstrated better results than the others.

The approaches involving user input may produce varying results depending on operator proficiency. To mitigate this, the points that experimenters need to input were clearly defined to ensure consistent outcomes regardless of skill level.

The marginal fit of dental prostheses is clinically significant,[1,2,40] particularly in digital dentistry. Its quality depends on the operator's skill level, but accurate finish line extraction by manual processing is a repetitive and time-consuming job.[5] Researchers and companies are striving to address this challenge, but the low robustness of automated finish line extraction leads to inconsistent quality of the definitive dental prosthesis.[3,7,8] Low accuracy is not solvable if it relies on user input. However, fully automated deep learning methods offer a solution for producing results without user input.[26,27] Even with these approaches, the finish line placement process remains challenging in terms of accuracy. Although finish line extraction and tooth boundary extraction share similar technical aspects, many research groups have investigated tooth boundary extraction[6–8,10,14,17–20,24,25,27,41] compared with finish line extraction.[1–3] Consequently, further research is necessary to automate finish line extraction.

Regarding the Dentbird software program and the Dental System software program (Group IS), which used fully automated methods, the Dentbird software program showed statistically significantly more accurate results with smaller HD and CD values (P<.05) (Tables 2, 3).

Table 2. *P* values obtained from *t* tests comparing different pairs of the software programs in terms of Hausdorff distance and chamfer distance

	Tooth Type	Metric (*P*)	A-B	A-C	A-D	B-C	B-D	C-D
Group DS	Anterior (n=33)	HD	.961	.392	.329	.623	.423	.063
		CD	.984	.181	.975	.625	.971	.161
	Posterior (n=52)	HD	.325	.124	<.001***	.525	.004**	.011*
		CD	.312	.730	.007**	.066	<.001***	.023*
Group IS	Anterior (n=32)	HD	.029*	.387	.102	.046*	.460	.118
		CD	.044*	.175	.156	.157	.413	.836
	Posterior (n=51)	HD	<.001***	.002**	<.001***	.116	.711	.097
		CD	<.001***	.012*	.008**	.009**	.706	.061

A, Dentbird software program; B, Dental System software program; C, DentalCAD software program; D, Medit software program
CD, chamfer distance; HD, Hausdorff distance
Significant differences: *P<.05; **P<.01; ***P<.001

Table 3. Mean and standard deviations of metric values

	Tooth Type	Metric		Dentbird	Dental System	DentalCAD	Medit
Group DS	Anterior (n=33)	HD [mm]	Mean	0.208	0.206	0.191	0.236
			SD	0.082	0.175	0.080	0.138
		CD [a.u.]	Mean	0.017	0.017	0.013	0.017
			SD	0.014	0.042	0.008	0.014
	Posterior (n=52)	HD [mm]	Mean	0.181	0.197	0.199	0.244
			SD	0.065	0.141	0.097	0.134
		CD [a.u.]	Mean	0.010	0.009	0.011	0.017
			SD	0.007	0.013	0.006	0.017
Group IS	Anterior (n=32)	HD [mm]	Mean	0.428	0.549	0.467	0.518
			SD	0.187	0.265	0.238	0.258
		CD [a.u.]	Mean	0.089	0.133	0.113	0.117
			SD	0.093	0.115	0.109	0.093
	Posterior (n=51)	HD [mm]	Mean	0.543	0.712	0.635	0.694
			SD	0.274	0.325	0.332	0.372
		CD [a.u.]	Mean	0.089	0.170	0.124	0.160
			SD	0.076	0.157	0.131	0.209

CD, chamfer distance; HD, Hausdorff distance; SD, standard deviation

Table 4. Total number of acceptable and unacceptable preparations and thresholds with false-positive fraction, true-positive fraction, and maximum classification accuracy

	Preparations			Metric (FPF, TPF, CA)	
	Total	Acceptable (Positive)	Unacceptable (Negative)	HD [mm]	CD [a.u.]
Group DS	340	305	35	0.366 (.29,.97,.94)	0.026 (.37,.96,.93)
Group IS	332	194	138	0.567 (.13,.89,.88)	0.100 (.14,.90,.88)

CA, classification accuracy; CD, chamfer distance; FPF, false-positive fraction; HD, Hausdorff distance; TPF, true-positive fraction
Thresholds for each metric selected to achieve maximum CA.

Additionally, the Dentbird software program demonstrated lower standard deviation values, indicating more consistent accuracy across preparations (Table 3). This approach, which integrates deep learning and CAD-based methods, has the potential to attain high robustness and accuracy without user input.

The HD focuses on this maximum distance and disregards the remaining point locations. In contrast, the CD reflects the overall similarity between the 2-point sets without being affected by a single point (Fig. 5B). Therefore, both the HD and CD should be considered to evaluate their similarity with reference data. In terms of the HD and CD metrics, the Dentbird software program exhibited the best performance, with statistically significant differences ($P<.05$) (Fig. 3).

The Dental System software program and the DentalCAD software program require the user to choose a method depending on the margin type. The method of the Medit software program is assumed to be determined by the convexity of the surface on a point selected by the user. When both concave and convex regions coexist (Fig. 6), the Dentbird software program demonstrated robust results independent of the convexity of the surface, as it used deep learning to train the data, including complex convexity preparations. In contrast, the others predefined the convexity of the surface, making it impossible to accurately determine the finish line.

Clinically acceptable threshold values for HD and CD were established (Table 4). The determination of a reasonable threshold value is essential and requires the involvement of dental laboratory technicians to assess the clinical acceptability of finish lines. Even if the measured metric shows it accurately, if the position of the finish line in relation to the preparation border in the axial and transversal directions is defined as a shorter imprecise fit, it can lead to plaque accumulation and the exposure of vulnerable dentin, resulting in dental caries.

In Group DS, all average CD and HD values from each software program met the threshold values. However, in Group IS, only the Dentbird software program exhibited average values of both the CD and HD that were lower than the threshold values, whereas the others did not meet these criteria (Table 3). In other words, the Dentbird software program can be considered clinically acceptable for both intraoral scans and desktop scans of casts trimmed for improved finish line visibility, while the others are clinically acceptable for desktop scans.

In Group DS, no software program exhibited an exceptional performance, and the results were generally similar. However, in Group IS, the Dentbird software program produced the best results (Fig. 3). This finding suggests that the hybrid method is suitable for finish lines on concave abutment regions, which are commonly found apical to the gingival margin and typically acquired using intraoral scans. Thus, given the growing use of intraoral scanning, the hybrid method plays an important role in chairside CAD.

Limitations of the study included that, in the hybrid method, the preparation tooth detector cannot detect natural tooth abutments in difficult situations involving multiple or minimally invasive preparations. This failure was observed in 12 (7%) of the 168 preparations. This method resolves the failure with user input. A possible approach to reduce failures is to provide a wider range of preparations in the training process for the preparation of tooth detectors. In addition, the imbalance between the number of acceptable and unacceptable preparations in Group DS may impact the reliability of the threshold values and ROC analysis. The ratio between acceptable and unacceptable preparations was approximately 9:1. This observation was anticipated because of the explicit and identifiable extraction of the finish line for Group DS using algorithms, leading to a higher proportion of acceptable preparations.

CONCLUSIONS

Based on the findings of this study, the following conclusions were drawn:

1. The hybrid method used by Dentbird software program outperformed existing commercial software solutions, particularly in its ability to accurately extract finish lines from intraoral scans as automated initial results. This implies a higher probability of being able to use these automated results with minimal or no additional adjustments.
2. Based on a hybrid method of deep learning and CAD approaches, robust and accurate finish line extraction can be achieved, leading to notable advantages in the field of digital dentistry.

DATA AVAILABILITY STATEMENT

Regrettably, we do not intend to disclose the data used in this study as a result of concerns related to confidentiality.

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CRediT authorship contribution statement

Jinhyeok Choi: Contributed to the study conception and design, as well as data analysis and interpretation. **Junseong Ahn**: Played a crucial role in the acquisition, analysis, and interpretation of data, and contributed to the drafting and review of the manuscript for important intellectual content. **Ji-Man Park**: Provided supervision and clinical expertise in the study, and granted final approval for the version to be published.

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